Ex:1

Basic Networking commands in Linux and Windows operating systems.

AIM:

To study and execute the basic networking commands in Windows and Linux operating systems for viewing network configuration, testing connectivity, checking routing information, monitoring network status, and troubleshooting network related issues.

Software Requirements:

* Windows 10/11 Linux (Ubuntu 20.04/22.04)
* Command Prompt (Windows).
* Terminal (Linux)
* Active Network connection (LAN/Wi-Fi).
* ipconfig/ifconfig/ip - Displays network configuration
* ping - Tests connectivity to a remote host.
* tracert/traceroute - Displays the path packets take to reach a destination.
* nslookup - Resolve domain names to IP address.
* netstat - Displays active network connection and and listening ports.
* arp - Displays the ARP cache.
* hostname - Displays the system hostname
* route - Displays or modifies the routing table.

Output:

I.O : ipconfig

O.P: Windows IP Configuration

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :

Link-local IPv6 Address . . . . . : Fe80::Fod1:3b:

IPv4 Address . . . . . : 192.168.56.1

Subnet Mask . . . . . : 255.255.255.0

Default gateway . . . . . :

Wireless LAN adapter local Area Connection* 1:

Media state . . . . . : Media disconnected

Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected

Connection-specific DNS suffix:

Wireless LAN adapter Wi-Fi:

Connection-specific DNS suffix :

Link Local IPv6 Address . . . : fe80::C43a:5045:94db:4103%8

IPv4 address . . . . . : 172.30.0.35

Subnet Mask . . . . . : 255.255.224.0

Default gateway : . . . . . : 172.30.0.1

Ethernet adapter Bluetooth Network Connection:

Media State . . . . : Media disconnected.

Connection-specific DNS suffix :

Ethernet adapter Ethernet:

Media State . . . . : Media disconnected

Connection-specific DNS Suffix . :

WINDOWS commands:

step 1: Open command Prompt.

1) Click start
2) Type CMD
3) Click command prompt.

Command:

ipconfig

Purpose.

Displays

* IP address
* Subnet Mask.
* Default gateway.

Step3: Display Detailed Network information.

Command

ipconfig /all

shows:

* MAC Address
* DHCP Server.
* DNS server
* lease information
* Adapter Details.

tep 4: Test Network Connectivity.

ommand

ping google.com.

I.P : IPconfig /all

Windows IP configuration

Host Name . . . . : LAPTOP-9MH74FT6

Primary DNS suffix . . . :

Node Type . . . . . : Mixed

IP Routing Enabled . . . : No

WINS Proxy Enabled . . . : No

Ethernet adapter Ethernet 2 :

Connection-Specific DNS suffix . :

Description . . . . : Virtual Box Host-Only Ethernet Adapter

Physical address . . . : 0A-00-27-00-00-10

DHCP enabled . . . . . : No

Autoconfiguration Enabled : Yes

Link-Local IPv6 Address . . : fe80::fod1:3b:da2e:fb22%16(Pr

IPV4 address . . . . . : 192.168.56.1 (Preffered)

Subnet Mask . . . . . : 255.255.255.0

Default gateway . . . . . :

DHCPV6 IAID . . . . . : 688521255

DHCPv6 Client DUID . . . : 00-01-00-01-2E-FC-C2-2C-BC-FC-E7-02-11-45

Net BIOS over Tcpip . . . : Enabled.

Wireless LAN adapter local Area Connection * 1 :

Media State . . . . : Media disconnected

Connection-specific DNS suffix :

Description . . . . : Microsoft Wifi Direct Virtual Adapter

Physical Address . . . : 70-15-FB-12-70-BA

DHCP Enabled . . . : Yes

Autoconfiguration Enabled : Yes.

Wireless LAN adapter Local Area Connection* 2 :

Media State . . . : Media disconnected

Connection-specific DNS suffix . :

Description . . . . : Microsoft Wifi Direct virtual adapter #2.

Physical Address . . . : 72-15-FB-12-70-B9

DHCP Enabled . . . . : No

Purpose:

Checks whether the destination is reachable.

Step 5: Trace Route to Destination.

Command

tracert google.com

Purpose:

Displays every router (hop) between your computer and destination.

Step 6: Display DNS information.

Command:

nslookup google.com

Purpose:

Displays

* DNS server
* IP address of website.

Stop

Auto configuration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS suffix . :

Description . . . . . : Intel (R) Wi-Fi 6E AX211 160MHz

Physical Address . . . . : 70-15-FB-12-70-B9

DHCP Enabled . . . . . : No

Auto Configuration enabled . . : Yes

I/P: ping google.com

O/P:

Pinging google.com [142.250.66.14] with 32 bytes of data:

Reply from 142.250.66.14: bytes=32 time=17ms TTL=118

Reply from 142.250.66.14: bytes=32 time=9ms TTL=118

Reply from 142.250.66.14: bytes=32 Time=13ms TTL=118

Reply from 142.250.66.14: bytes=32 time=25ms TTL=118

Ping statistics for 142.250.66.14:

Packets: Sent = 4, Received = 4, Lost = 0 (0% Loss),

Approximate round trip times in milliseconds:

Minimum = 9ms, Maximum = 25ms, Average = 16ms.

I/P: tracert google.com

O/P:

Tracing route to google.com [142.250.66.14]

over a maximum of 30 hops:

1	1ms	1ms	1ms	172.30.0.1
2	2ms	2ms	2ms	137.97.163.129
3	*	*	*	Request timed out
4	50ms	6ms	6ms	172.16.12.64
5	5ms	5ms	5ms	172.16.12.64
6	5ms	5ms	5ms	209.85.175.48
7	5ms	5ms	6ms	72.14.234.9
8	6ms	6ms	8ms	142.251.60.187
9	7ms	6ms	6ms	bom07s35-in-f14.1e100. [142.250.66.14]

Trace Complete.

Step 7: Display Routing Table

Command:

route print

Purpose:

Displays:

* Network Routes
* Gateways
* Destination Networks.

Step 8: Display Active Connections.

Command:

netstat

Purpose:

Displays:

* Active TCP connections
* Listening Ports.

I/P: nslookup google.com

O/P:

Server: dns.google

Address: 8.8.8.8

Non-authoritative answer:

Name: google.com

Addresses: 2404:6800:4007:808::200e

142.250.66.14

I/P: Route print

O/P:

Interface List

16...0a 00 27 00 00 10 ..... VirtualBox Host-only Ethernet Adapter

14...70 15 fb 12 70 ba ..... Microsoft Wifi Direct Virtual adapter

19...72 15 fb 12 70 b9 ..... Microsoft Wifi Direct Virtual Adapter #

8...70 15 fb 12 70 b9 ..... Intel(R) Wifi 6E AX211 160MHz

IPv4 Route Table

Active Routes:

Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	172.30.0.1	172.30.0.35	
127.0.0.0	255.0.0.0	On-Link	127.0.0.1	33
127.0.0.1	255.255.255.255	On-Link	127.0.0.1	33
127.255.255.255	255.255.255.255	On-Link	127.0.0.1	33

Persistent Routes:

None

IPv6 Route Table

Active Routes:

If	Metric	Network destination	Gateway
1	331	::1/128	On-link
16	291	fe80::/64	On-link
8	331	fe80::/64	On-link
8	331	fe80::c439:5045:94db:4103/128	On-link

Persistent Routes:

None

Step 9: Display Network Statistics.

Command:

netstat -a

shows:

* All ports
* TCP connections
* UDP ports

Step 10: Display MAC Address.

Command

getmac

Purpose:

Displays MAC address of the computer.

I/P: netstat

O/P:

Active Connections

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:1042	Kubernetes:53223	Established
TCP	127.0.0.1:1042	Kubernetes:61980	Established
TCP	127.0.0.1:19575	Kubernetes:49338	TIME_WAIT
TCP	127.0.0.1:19575	Kubernetes:65476	Time_wait
TCP	127.0.0.1:49694	Kubernetes:49693	established
TCP	127.0.0.1:49735	Kubernetes:13031	established

I/P: netstat -a

O/P:

Active connections:

Proto	Local Address	Foreign address	State
TCP	0.0.0.0:135	LAPTOP-9MH7FT6:0	Listening
TCP	0.0.0.0:49729	LAPTOP-9MH7FT6:0	Listening

I/P: getmac

O/P:

Physical address	Transport name
BC-FC-E7-0C-11-45	Media disconnected
70-15-FB-12-70-D9	\Device\Tcpip_{4C46E192-BF1A-4TAB-A946-BC6F4FDCB}

I/P: hostname

O/P: LAPTOP-9MH74FT6

I/P: arp -a

O/P: Interface: 172.30.0.35 ---0x8

Step 11: Check Hostname

Command:

hostname

Purpose:

Display computer name.

Step 12: View ARP Table

Command:

arp -a

Purpose:

Displays IP-to-MAC address mappings.

Internet Address	Physical address	Type
172.30.0.1	CC-82-7f-30-b5-03	dynamic
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static

Interface : 192.168.56.1 --- 0x10

Internet Address	Physical address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static

Linux Networking Commands.

Step 1: Open Terminal

Power:
Ctrl + Alt + T

OR
Search for terminal

Step 2: Display IP Address.

Command:
ip addr

OR
ifconfig
(If installed)

Purpose
Displays.

Step 3: Display Routing table.

Command:
ip route

Purpose

Shows
* Default gate way
* Routing table.

Step 4: Display Host name

Command:

hostname

Purpose:

Displays system hostname

Step 5: Test Connectivity.

Command:

Ping google.com

stop using:

ctrl + C

Step 6: Trace Network Route

Command:

traceroute google.com

if not installed:

sudo apt install traceroute

step 7: DNS look up

Command:

nslookup google.com

if not installed:

sudo apt install dnsutils

step 8: Display Network statistics.

Command:

netstat -a

OR

ss -a

step 9: Display ARP table.

Command:

arp -a

OR

ip neigh

step 10: Display MAC Address

command:

ip link

OR

ifconfig

Look for

link / ether

step 11: Display DNS configuration

Command:

cat /etc/resolv.conf

Shows

nameserver 8.8.8.8

step 12: Check interface status

Command:

ip link show

Displays:

* UP / DOWN status
* interface name.

step 13: Display listening ports

Command:

ss -tulpn

Purpose:

Displays:

* TCP ports
* UDP ports
* Listening services

Step 14: Check Internet connectivity.

Command:

curl google.com

OR

wget google.com

ult:

Basic network commands were succesfully ex
both Linux and windows.